

Verdalis — decarbonising industrial heat

The Problem

Industrial heat is responsible for roughly a fifth of global energy-related emissions. It is the hardest part of the energy transition and the least addressed: electricity generation has a decarbonisation path, road transport has one, and industrial process heat above 200°C largely does not.

Cement, glass, ceramics, chemicals and food processing all depend on it. None of them can electrify with existing technology at a cost their margins survive.

This is the largest unsolved problem in climate and it is worth tens of billions a year.

Our Solution

Verdalis is developing a high-temperature thermal storage and delivery system that charges on cheap renewable electricity and discharges heat on demand at process temperature.

The concept is to store energy in a low-cost solid medium during periods of negative or near-zero electricity prices, which now occur for over a thousand hours a year in several European markets, and deliver it as continuous process heat.

We are at the design stage. The thermodynamic modelling is complete and a bench-scale demonstration is planned for next year.

The Market

Industrial heat represents an estimated €180B of annual energy spend across Europe alone. Carbon pricing under the EU emissions trading system reached €94 a tonne last year, and the free allocation that has shielded heavy industry is being phased out between now and 2034.

That phase-out is a schedule, not a forecast. Every affected plant will need a decarbonisation answer on a known date, and most of them do not have one. The urgency is not something we are asserting — it is legislated.

Four of our target industries have announced net-zero commitments with interim 2030 targets they currently have no means of meeting.

Competition

Three companies work on thermal storage for industrial heat, two of them with operating pilots. Electrode boilers and industrial heat pumps address lower temperature ranges. Hydrogen is the incumbent alternative and is expensive.

We have not yet identified what distinguishes our approach from the two companies with operating pilots, beyond a projected cost per kilowatt-hour of storage that our modelling suggests and their published figures do not yet match.

The Team

Thomas Aldridge spent six years in energy consulting, advising industrial clients on decarbonisation. Sophie Marchand has a doctorate in thermal engineering and joined full time four months ago.

Four people, two of them part time.

Traction

- thermodynamic model complete
- one industrial partner has signed a memorandum of understanding to host a future pilot
- two grant applications submitted, neither decided
- no prototype, no experimental data, no revenue

Business Model

We expect to sell heat as a service under long-term contracts, with the capital carried by project finance rather than by the customer.

The pricing depends on the cost per kilowatt-hour of installed storage, which we will not know until we have built something.

Go-To-Market

Not yet defined.

Funding

We are raising €4M to build the bench-scale demonstrator and produce the first experimental data.